

Research Article

Wearable Technology for Elderly Care: Integrating Health Monitoring and Emergency Alerts

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With the growing elderly population, ensuring consistent and reliable healthcare monitoring has become increasingly important. This research introduces an innovative solution in the form of a smart glove designed specifically for elderly care. The glove is equipped with advanced features such as fall detection, emergency messaging via hand gestures, and vital signs monitoring, including oxygen saturation (SpO₂) and heart rate. The smart glove integrates the MPU6050 and MAX30100 sensors with Arduino and ESP8266 modules to achieve these functionalities. Fall detection is accurately identified by the MPU6050 sensor, while the MAX30100 sensor monitors SpO₂ and heart rate. Hand gestures are recognized and interpreted for sending emergency messages. Data from these sensors are transmitted through IFTTT, enabling alerts and notifications via SMS, email, and a secured website, thus facilitating real-time remote monitoring and communication. This setup provides a comprehensive and reliable method for monitoring the health and safety of elderly individuals, ensuring prompt response and care in emergencies. The proposed system demonstrates significant potential in enhancing the quality of life for elderly individuals by offering a smart, integrated, and user-friendly healthcare solution.

Keywords: elderly care; fall detection; heart rate; Internet of Things; smart glove; SpO₂

1. Introduction

The rapid advancement of technology is driving the shift toward a modern civilization, with the Internet of Things (IoT) being one of the latest innovations. IoT facilitates the efficient exchange of large volumes of data, enabling remote monitoring and control of various objects. Its applications are especially prevalent in the medical field. This study presents a smart glove powered by IoT, specifically designed for the elderly. The glove addresses key eldercare challenges by detecting falls, monitoring vital signs like heart rate and SpO₂, and sending emergency alerts through hand gestures.

According to the World Health Organization (WHO), falls are the second leading cause of accidental or

unintentional fatalities worldwide [1]. Approximately 6 million people are affected by fatal falls, with a significant 80% of these incidents occurring in low- and middle-income countries. Among the elderly population, an alarming 37.3 million cases involve falls severe enough to require medical intervention. The lack of comprehensive healthcare infrastructure and specialized medical professionals in remote and rural regions underscores the urgent need for affordable, rapid, and effective strategies to detect falls promptly. These strategies could include both resource-efficient infrastructure improvements and autonomous, expert-driven systems. Focusing on the fall detection feature, when an elderly individual experiences a fall, the sensor integrated within the smart glove immediately identifies the incident.

These data are then sent to a microcontroller, which employs a Wi-Fi module to transmit an alert message and email to a mobile device via a web-based service.

A major concern is the rising aging population, a trend observed over the past decade and anticipated to continue for the next 20 years due to increased life expectancy and decreasing fertility rates [2, 3]. As individuals grow older, their bodies become more vulnerable to various age-related conditions such as hypertension, heart failure, coronary artery disease, high blood pressure, and diabetes [4]. Monitoring key vital signs—heart rate, blood pressure, respiration rate, temperature, and peripheral capillary oxygen saturation (SpO_2)—is crucial, with SpO_2 being particularly significant as inadequate oxygen levels can lead to irreversible damage to vital organs [5]. Low SpO_2 readings are associated with higher risks of morbidity and mortality. To address this issue, a wearable device has been proposed to monitor SpO_2 and heart rate in real time for the elderly, thereby enhancing emergency readiness related to oxygen and heart rate. Specifically, a smart glove has been developed to track these vital health parameters using sensors and a microcontroller. The collected data are wirelessly transmitted and displayed in real time on websites, Google Sheets, and an LCD screen, thereby improving emergency preparedness and response.

Effective communication is crucial to our daily lives, especially in a world where a significant portion of the population includes elderly and disabled individuals who may face difficulties with speaking and mobility. In recent decades, innovations such as electric wheelchairs and hand signals have significantly enhanced the quality of life for those with physical impairments [6–8]. These technologies offer valuable mobility solutions for individuals with both upper and lower body disabilities. Research indicates that 466 million people worldwide experience profound deafness and limited abilities, constituting 6% of the global population [9]. Of these, 75 million individuals require wheelchairs, which equates to about 1% of the world's population. To address the communication challenges faced by these individuals, a practical solution has been developed: a hand glove equipped with a specialized emergency alert sensor. This sensor enables users to send messages through directional hand gestures—forward, backward, right, and left—similar to traditional hand signals. These gestures activate the transmission of messages via email and mobile devices, and also display the messages on an LCD screen, thus facilitating effective communication and emergency alerting.

A considerable amount of research has focused on detecting and preventing falls among the elderly, as seen in studies [10, 11]. Various approaches have been explored to detect falls, involving a range of sensors and methods [12–14]. However, the use of multiple sensors can increase costs and add complexity for elderly users. Much of the research in this area has concentrated on platforms such as Arduino and MSP430 [11, 12, 15, 16]. Relevant literature includes studies on processing SpO_2 and heart rate data [17, 18], the advancement of IoT wearable health devices [19, 20], and the use of the MAX30100 sensor [21, 22]. In addition, the application of IoT in healthcare, including

electrocardiogram (ECG) monitoring [23] and blood pressure measurement [24], has been extensively discussed. Some reports have also explored integrating multiple health parameters into a single system [25]. Recent research efforts have increasingly focused on aiding individuals with disabilities. For example, Li et al. [26] developed an AI-based intelligent robot to assist autistic children with communication and mobility. Gomez-Donoso et al. introduced a multisensor wheelchair designed for people with disabilities, while Li et al. [27] proposed a head gesture device for wirelessly controlling wheelchairs, addressing the limitations of traditional joystick controls. Current research [28–30] on robotic wheelchairs highlights various challenges [31–33]. Wan et al. [34] presented a wearable gesture-controlled robot that uses motion sensors and machine learning for accurate gesture recognition. An intelligent wheelchair [35] designed for quadriplegic patients incorporates sensors for self-navigation and obstacle detection. Sheares et al. [36] proposed a smart wheelchair with navigation capabilities that utilizes head gestures and an accelerometer sensor for controlled movement. The study [37] of wearable technologies for elderly care focuses on integrating real-time health monitoring and emergency alert systems into user-friendly devices. This includes using sensors like the MPU6050 for fall detection and the MAX30100 for vital signs monitoring (SpO_2 and heart rate), ensuring that elderly individuals receive prompt care through automated alerts and communication.

This study introduces an IoT-based smart hand glove, which acts as a multifunctional solution to three key challenges encountered by the elderly. Rigorous testing was conducted to guarantee its reliable functionality, utilizing diverse platforms to exhibit results and address problems. In situations where the elderly fall or require assistance with tasks like going to the bathroom or getting water, the hand glove sends instant email notifications for aid, functioning seamlessly even without an internet connection, and dispatching alerts to mobile devices. Furthermore, a purpose-built website has been established to showcase essential health metrics like heart rate and oxygen saturation, complemented by the availability of these data on a Google Sheet. This inventive solution particularly advantages individuals residing in rural areas by negating the necessity to travel for checkups and diminishing the reliance on full-time caregiving services. This innovative system significantly reduces the financial burden associated with frequent checkups, transportation costs to attend appointments, and the hiring of a caretaker. The ultimate aim is to furnish an uncomplicated, financially viable remedy to the three pivotal hurdles confronted by the elderly, thereby guaranteeing outcomes of commendable quality. The main objective of this work is to offer a straightforward and cost-effective solution to three significant challenges faced by the elderly, ensuring high-quality outcomes.

2. Materials and Method

The purpose of this system is to design a smart hand glove through which various crucial health factors of elderly

people can be measured. It also detects falls and sends emergency messages through hand gestures. In this section, an elaborate explanation of the methodologies and materials utilized in this study is presented. The first subsection introduces the block diagram of the proposed system.

2.1. Block Diagram. The block diagram of the proposed system is depicted in Figure 1. The identification of falls among senior citizens is executed through the utilization of the mpu6050 sensor. Then, the microcontroller and Wi-Fi module collaborate to disseminate this vital information to a predetermined recipient's mobile device, thereby facilitating timely SMS and email notifications. Furthermore, an integrated buzzer system has been ingeniously integrated, primed to trigger in the regrettable scenario of the falling of the elderly individual. For initiating an emergency message, the patient will perform a specific hand gesture in the desired direction. This action will then be detected by the mpu6050 sensor, triggering the transmission of notifications to a designated person's mobile device through SMS and email. Utilizing the Max30100 sensor enables the monitoring of heart rate and oxygen saturation (SpO_2) in the elderly. This sensor collects the essential data, which is then transmitted to both a dedicated website and a Google Sheet through the microcontroller. Subsequently, these data are forwarded via the Wi-Fi module for transmission. Furthermore, the data can be conveniently visualized on the LCD screen, which offers a user-friendly interface for clear information.

The flowchart in Figure 2 outlines a system that monitors falls and hand movements. The process begins with the initialization of the MPU6050 sensor, which captures motion data. A Wi-Fi connection is then established to enable data transmission. The system continuously checks for falls and hand movements in a loop. Upon detecting a fall, the system triggers a multistep response: an audible alert sounds, an LCD displays a visual alert, and notifications are sent to caretakers via the Internet. For hand movements, the data are displayed on the LCD and shared online. This system ensures constant monitoring and timely responses to incidents.

Table 1 summarizes the components used in the wearable system, including a hand glove, sensors, microcontroller, display, and connectivity modules. Each component, such as the MPU6050 for motion tracking and MAX30100 for health monitoring, plays a crucial role in the system's functionality. Together, these parts enable real-time data acquisition, processing, and display.

This research project is structured into three key sections: fall detection, hand gesture communication, and monitoring of oxygen saturation (SpO_2) and heart rate levels. The system employs an Arduino Uno microcontroller to interface with a Wi-Fi network, processing input data from a specially designed hand glove. The collected data are transmitted to the authorized caregiver via email and SMS. For real-time visualization, the results are also displayed on an LCD screen. The subsequent sections detail the methodology for implementing each of these functions, and Figure 3 shows the partial circuit diagram of the system.

2.2. Fall Detection. To accomplish the task, the MPU6050 sensor, which integrates both a gyroscope and an accelerometer, is utilized. This sensor module features a three-axis accelerometer and a three-axis gyroscope. The gyroscope determines orientation, while the accelerometer provides data on angular parameters along the X, Y, and Z axes. To detect a fall, the magnitude of acceleration is compared to a predefined threshold. The fall detection signal from the MPU6050 sensor is transmitted to an Arduino UNO microcontroller connected to the sensor. The Arduino UNO then sends a signal to an ESP8266, which processes the information and forwards it to IFTTT.

IFTTT, which stands for "If This Then That," operates over Wi-Fi to send messages through its web service. It utilizes conditional statements, known as applets, to automate tasks. In the event of a fall, IFTTT applets trigger notifications, sending both email and SMS alerts to caregivers. In addition, the fall detection message is displayed on an LCD screen, and a buzzer sounds to alert when a fall is detected. This system provides a straightforward approach to monitoring elderly individuals.

2.3. Emergency Message With Hand Gesture. The MPU6050 sensor facilitates the transmission of emergency messages through hand gestures. Functioning as both an accelerometer and a gyroscope, it measures rotational movement across three axes, detects gravitational acceleration, and captures various motion parameters. Using the accelerometer capability of the MPU6050, elderly individuals can convey four critical messages by moving their hands in specific directions: forward, backward, right, and left. Figure 4 shows the smart glove in four different directions.

For example, a rightward hand motion sends a "Need Food" message, while a leftward wave communicates "Need Water." Moving the hand forward indicates "Emergency Help Coming," and a backward motion signals "Call Attender." The MPU6050 sensor transmits these directional signals to the ESP8266 when the hands move in any specified direction. The NodeMCU ESP8266 then connects to Wi-Fi and forwards the messages to the IFTTT web service. Through IFTTT, the elderly person's message is efficiently sent to the caregiver's phone via SMS and email.

2.4. SpO_2 and Heart Rate Monitoring. Firstly, a complete circuit is built to monitor both SpO_2 and heart rate. The MAX30100 sensor is employed to carry out the measurements. To facilitate this, an open-source customizable microcontroller board called Arduino Uno is linked to the Max30100 sensor. Concurrently, the NodeMCU ESP8266 microcontroller is utilized to establish a connection between the Arduino Uno and to gather data from elderly patients. It's important to note that the NodeMCU ESP8266 operates as a Wi-Fi module, enabling the transmission of data to the online platform.

The information gathered from the sensor can be accessed in two manners: one involves using a website, and the other entails utilizing Google Sheets. In addition, the collected data will also be visible on an LCD screen. The data

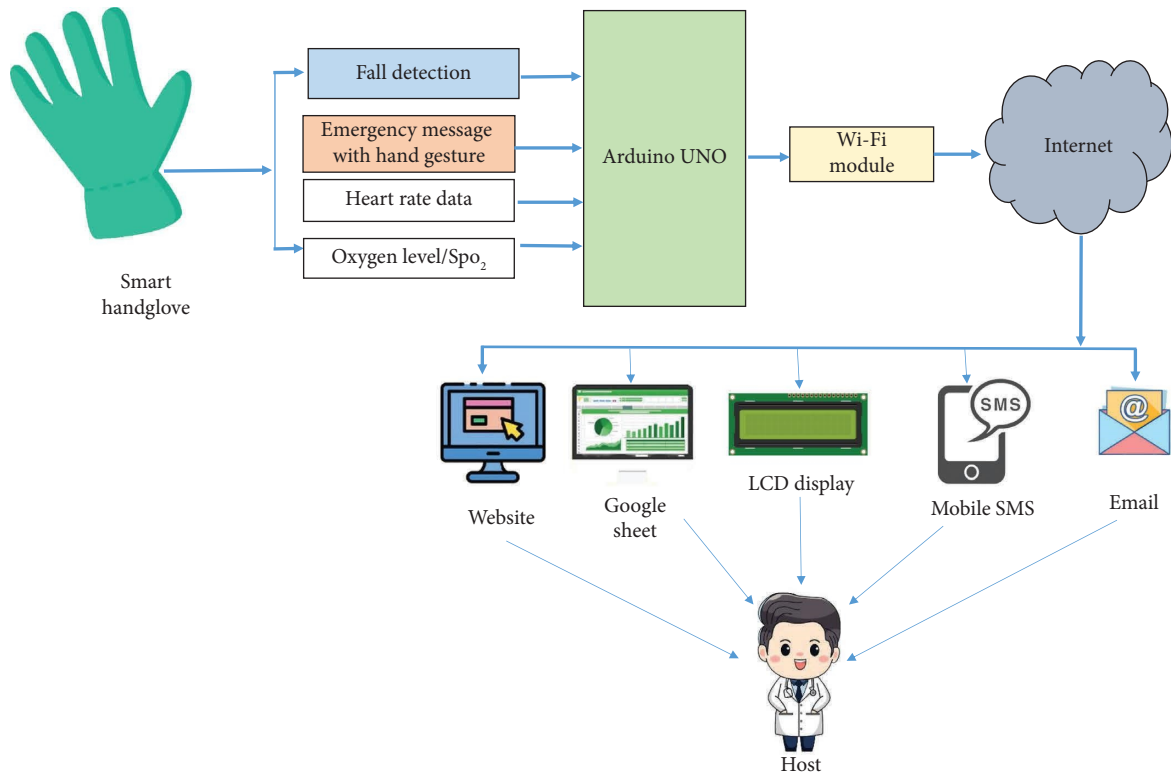


FIGURE 1: Block diagram of the proposed system.

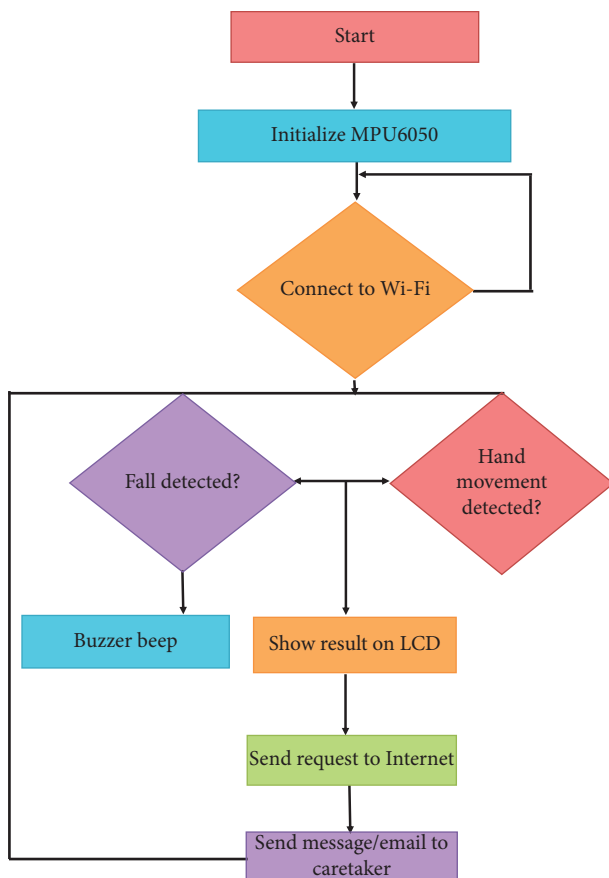


FIGURE 2: Flowchart of the system.

will be leveled within a Google Sheet, after which it will be directed to the website. This setup permits authorized individuals like doctors, caregivers, and family members of elderly individuals to view previous records of oxygen saturation (SpO2) and heart rate readings. The website is password protected, limiting entry to those mentioned above. This comprehensive system allows for real-time remote monitoring of the health status of elderly patients by medical caregivers.

2.5. Cost of Materials. The project's development and production focused on achieving a cost-effective and affordable price, as detailed in Table 2. By applying innovative design techniques, the total system cost is kept at 22.50 USD, ensuring an economical solution. The costs for acquiring the materials needed for the device construction are thoroughly described in Table 2. The goal was to set a competitive price of 22.50 USD, providing excellent value. Table 2 offers a comprehensive breakdown of the overall expenses.

Figure 5 illustrates the smart glove's integrated system, featuring sensors for fall detection, vital sign monitoring, and gesture-based emergency alerts. Data are processed via Arduino and transmitted through ESP8266 for remote notifications via IFTTT.

3. Result and Analysis

3.1. Output of the Smart Hand Glove. This section discusses the system developed for this research and summarizes the results achieved. The setup comprises an MPU6050, a heart

TABLE 1: Description of the components.

Components	Description
Hand glove	A hand glove is used to operate this system
Arduino UNO	The Arduino Uno is a microcontroller board that utilizes the Microchip ATmega328P microcontroller (MCU). It offers digital and analog input/output (I/O) pins—14 digital and 6 analog. This board is programmable using the Arduino IDE and can be powered through a USB cable connection accommodating 7 to 20 V voltages
MPU6050	The MPU6050 module functions as a motion tracking device containing a 3-axis accelerometer and 3-axis gyroscope, enabling measurement of acceleration, velocity, displacement, and other motion-related parameters for objects
MAX30100	The MAX30100 serves as a comprehensive solution for pulse oximetry and heart rate monitoring, integrating optimized optics, a photodetector, dual LEDs, and precise analog signal processing
ESP 8266	The ESP8266 is a microchip that integrates built-in TCP/IP networking software and operates as both a microcontroller and a Wi-Fi-enabled device
LCD display	LCDs, extensively used in electronics, visually convey information and status; the LCD 16×2 type specifically shows 2 lines with 16 characters
Bread board	A bread board serves as a platform for creating prototypes of electronic circuits
Buzzer	A buzzer is an auditory device that transforms audio patterns into sound signals
Switch	A push button switch is an apparatus utilized for managing an electrical circuit
Jumper wire	A jumper wire is an electrical cable utilized to link distant electric circuits in printed circuit boards

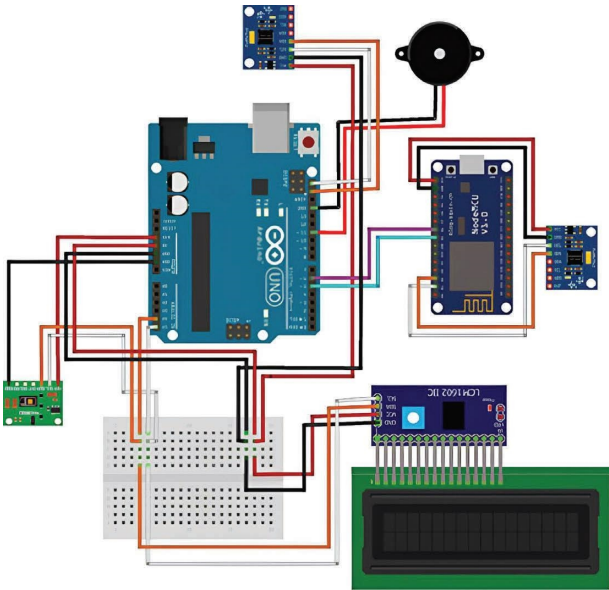


FIGURE 3: Partial circuit diagram of the system.

rate sensor, and an oximeter pulse sensor, all interfaced with an Arduino Uno microcontroller. The Arduino Uno, powered through a USB connection to a device, receives data from the sensors. These data are then transmitted to the Arduino and displayed on various platforms using a Wi-Fi module. It can be viewed on the Arduino IDE’s serial monitor, a website, received via email, sent as text messages to mobile devices, recorded in a Google Sheet, and shown on an LCD screen. Figure 5 illustrates a detailed diagram of the entire system.

In Figure 6, the Arduino IDE’s serial monitor displays information related to fall detection, with the data sourced from the MPU6050 sensor. In Figure 7, if an elderly person

experiences a fall, the MPU6050 sensor will detect it, and the message “Emergency Help Coming” will be displayed on the LCD screen, controlled by the Arduino Uno microcontroller. In addition, the LCD shows the output when different hand gestures are made to convey emergency messages. For instance, when elderly people move their hand glove to the left, it signals a need for water; when moved to the right, it indicates a request for food; and when moved forward, it signifies an urgent requirement. The MPU6050 sensor detects these gestures, prompting corresponding messages on the LCD screen.

Figure 8 shows that when an elderly person falls, the MPU6050 sensor detects it, and the fall event is communicated to a designated person’s mobile device via the ESP8266 Wi-Fi module. In Figure 9, it is demonstrated that the elderly person can send important messages to specific contacts by making directional hand gestures, with the IFTTT applet being used to send these emails.

In Figure 10, the heart rate and SpO₂ data are displayed on the Arduino IDE’s serial monitor, with the information gathered from the MAX30100 sensor. Figure 11 presents the heart rate and oxygen saturation (SpO₂) readings on the LCD screen.

A website has been established to enable specific individuals to remotely monitor crucial health metrics such as oxygen levels and heart rate, regardless of their physical distance. Figure 12 displays the login page of the website, which is fully secured with a password. Only the designated individual is permitted access, ensuring complete protection from unauthorized users. In Figure 13, a different patient dashboard is displayed. Figure 14 shows a patient’s health data for a full month. The datasheet for this includes reference ranges for both healthy and unhealthy conditions. Data collection for system validation will involve five groups of subjects:

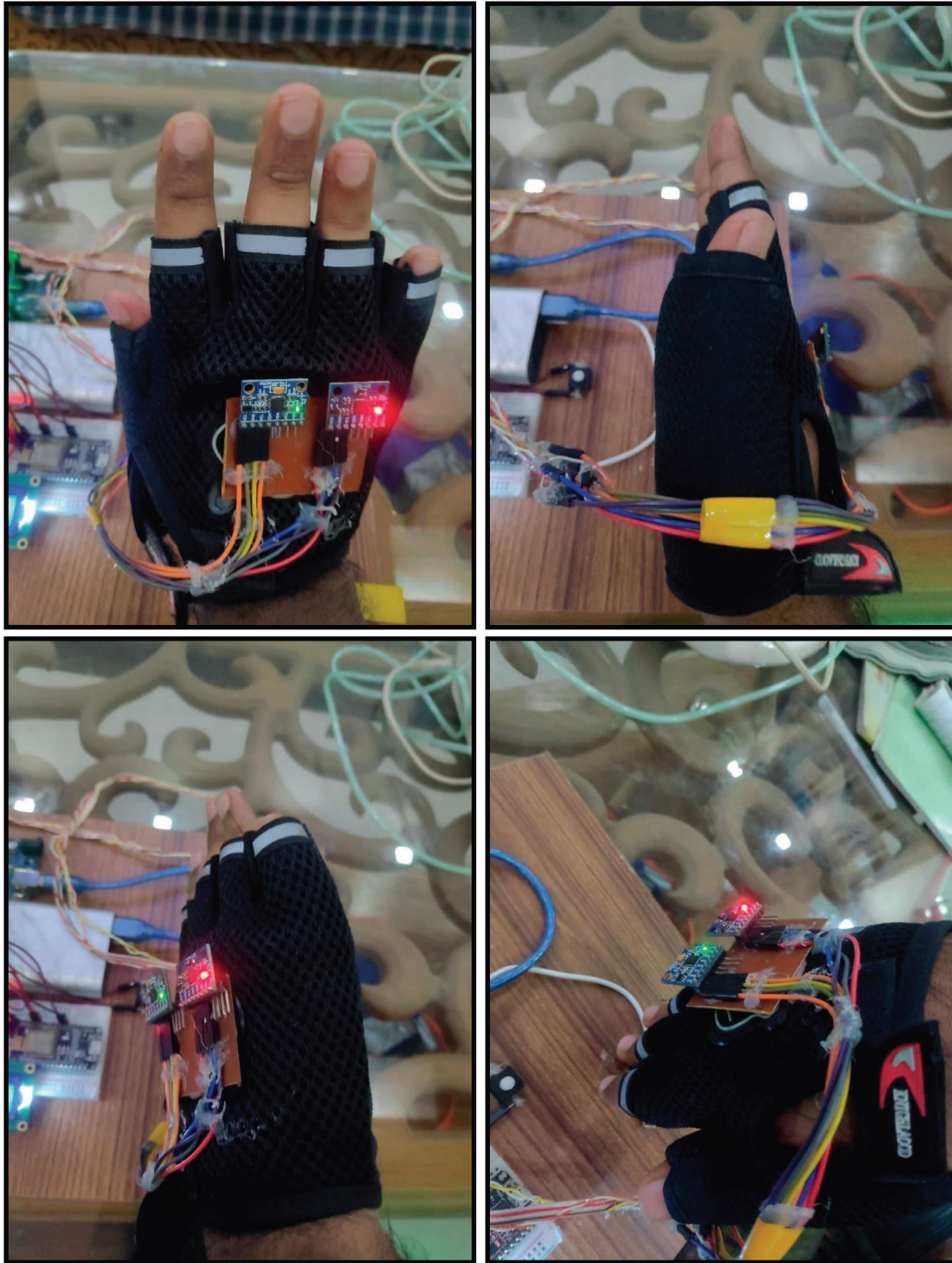


FIGURE 4: Moving the hand in four different directions with smart glove.

normal, abnormal, mild hypoxemia, moderate hypoxemia, and severe hypoxemia. These data will be used as a reference, in accordance with Health Regulation [5]. In Figure 15, it is demonstrated that a particular individual can observe the patient's health data remotely by utilizing Google Sheets.

3.2. Evaluation Metrics of the Hand Glove

3.2.1. Evaluation Metrics of the Fall Detection. The fall detection performance was assessed using three key metrics: precision, recall (sensitivity), and *F1*-Score. These metrics are calculated as follows:

TABLE 2: Cost of the components.

Component	Model	Quantity	Price (USD)
Microcontroller	Arduino UNO	1	9
Accelerometer gyroscope	MPU6050	1	2
Heart rate and oximeter pulse sensor	Max 30100	1	3
Wi-Fi module	ESP 8266	1	3
LCD display	16X2 I2C	1	2.50
Buzzer	AUD-00002	1	1
Switch		1	1
Hand glove		1	1.50
Bread board		1	1.50
Jumper wire		10 × 3	1
Total cost			22.50

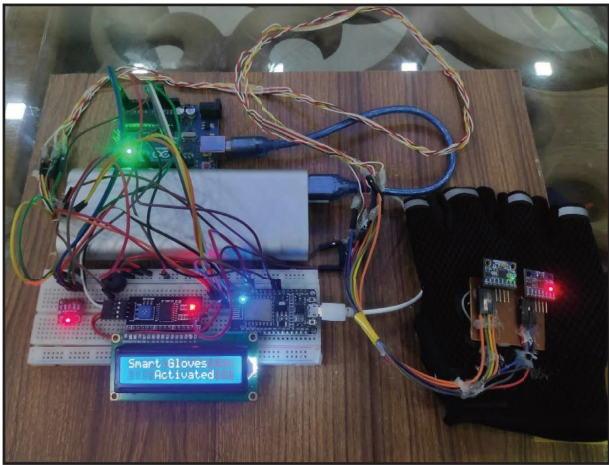


FIGURE 5: Total system model.

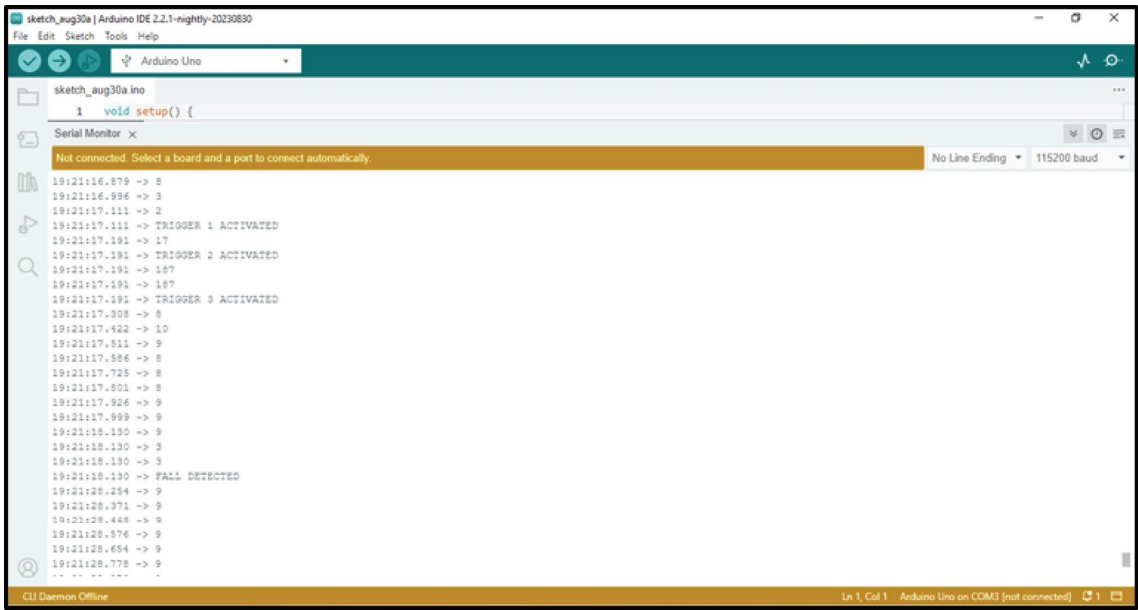


FIGURE 6: Diagram showing the measurement of fall detection in the serial monitor of Arduino IDE.

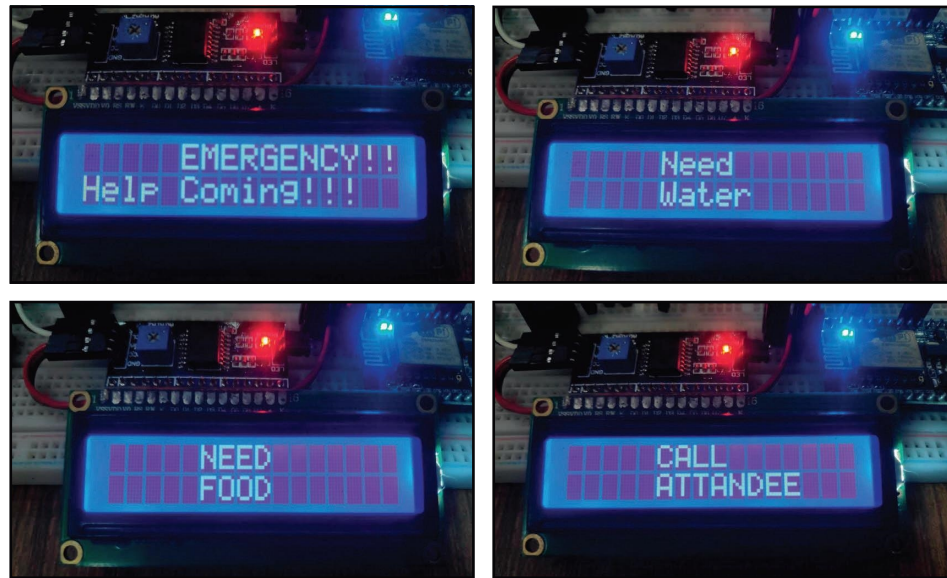


FIGURE 7: Displaying results on LCD after fall detection and emergency message after hand movement at different angles.

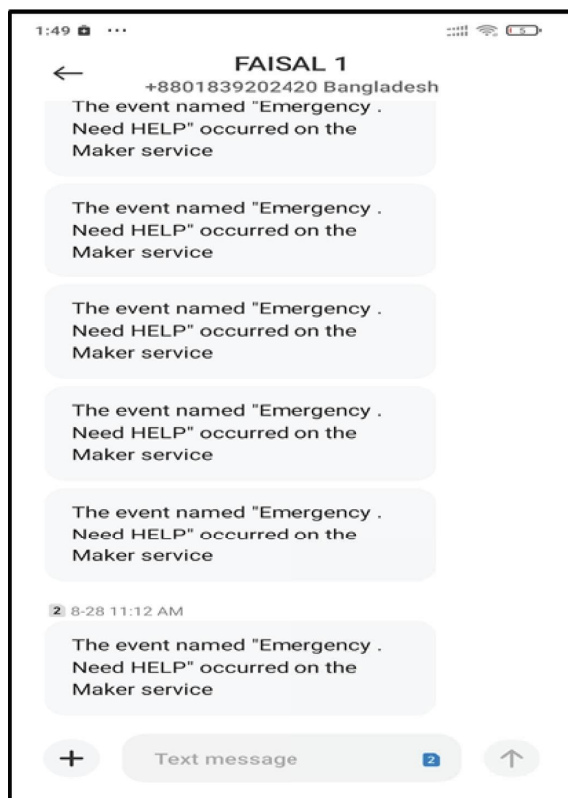


FIGURE 8: Diagram illustrating the process of sending an SMS to a mobile device following fall detection.

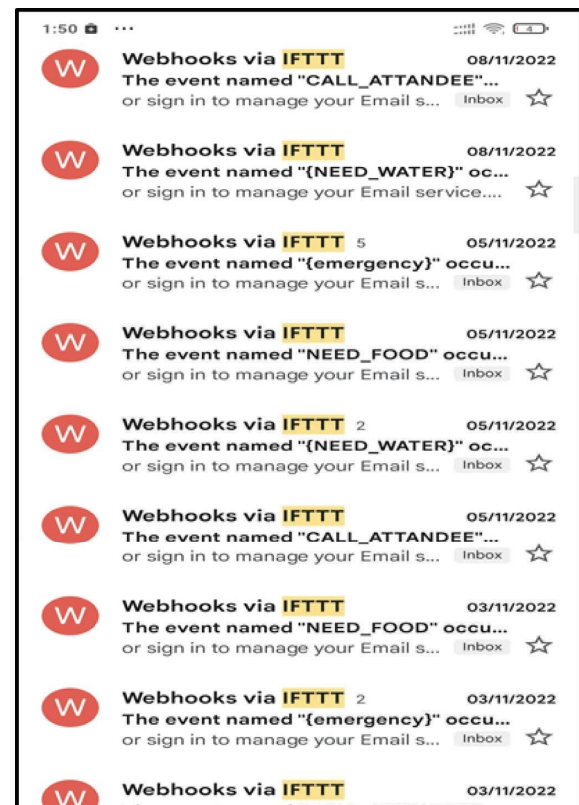


FIGURE 9: Diagram illustrating the process of sending an email triggered by a hand glove gesture.

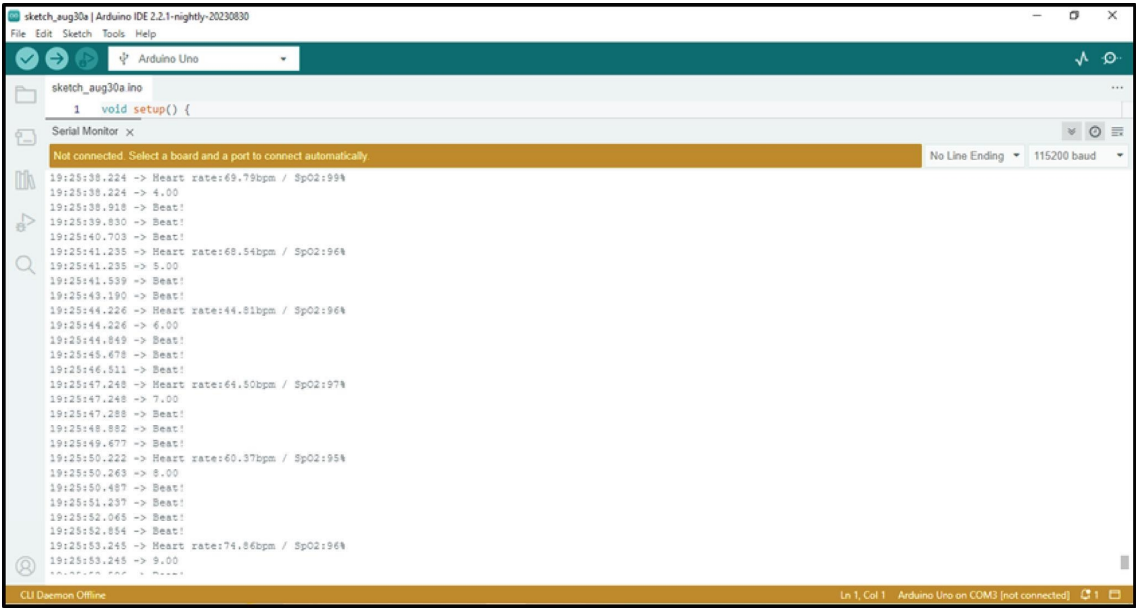


FIGURE 10: Diagram showing the measurement of heart rate and oxygen level in the serial monitor of Arduino IDE.

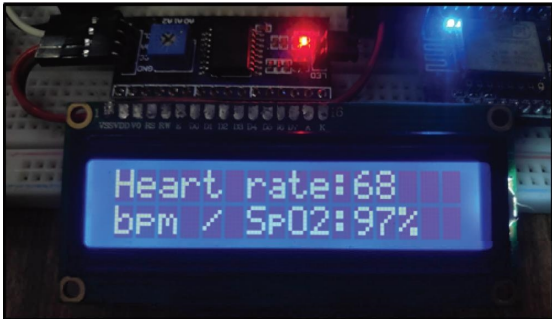


FIGURE 11: Heart rate and oxygen saturation (SpO₂).

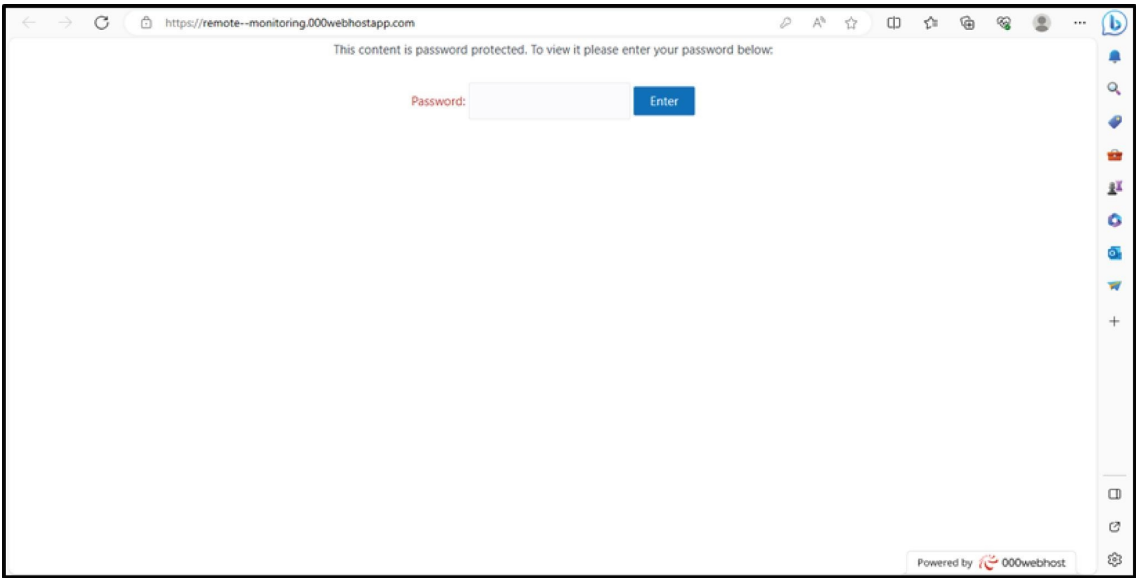


FIGURE 12: Login page of the website.

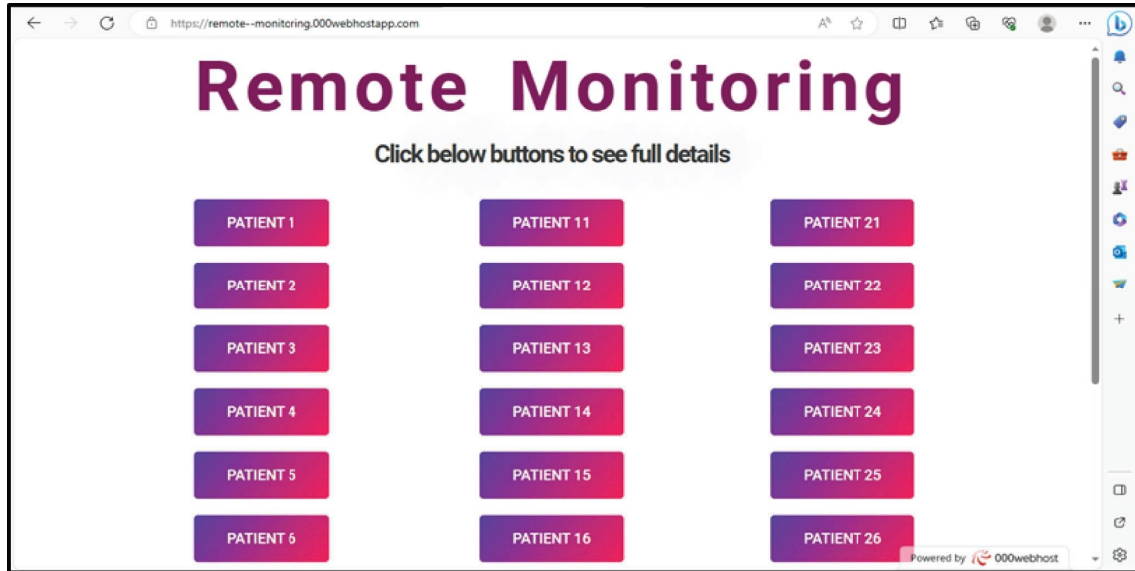


FIGURE 13: Patients dashboard on website.

ID	DATE	Heart rate(Bpm)	SpO2 (%)	Bpm Condition	SpO2 Condition
1	8/12/2022	0	0	Inactive	Inactive
2	8/12/2022	45	98	Abnormal	Normal
3	8/12/2022	38	93	Abnormal	Mild Hypoxemia
4	10/29/2022	50	60	Abnormal	Severe hypoxemia
5	10/29/2022	69	97	Normal	Normal
6	10/29/2022	77	97	Normal	Normal
7	10/29/2022	70	97	Normal	Normal
10	10/29/2022	80	205	Normal	Normal
11	10/29/2022	68	96	Normal	Normal
12	10/29/2022	75	215	Normal	Normal

FIGURE 14: Heart rate and SpO₂ information for the entire month.

Precision: It reflects the accuracy of positive predictions, indicating the proportion of detected falls that were actual falls:

$$\text{precision} = \frac{\text{true positives (TP)}}{\text{true positives (TP)} + \text{false positives (FP)}} \times 100. \quad (1)$$

Recall (sensitivity): It measures the system's ability to correctly identify all actual falls, representing the proportion of falls accurately detected.

$$\text{recall} = \frac{\text{true positives (TP)}}{\text{true positives (TP)} + \text{false negatives (FN)}} \times 100. \quad (2)$$

ID	DATE	Heart rate(Bpm)	SpO2 (%)	Bpm Condition	SpO2 Condition
1	8/12/2022	0	0	Inactive	Inactive
2	8/12/2022	45	98	Abnormal	Normal
3	8/12/2022	38	93	Abnormal	Mild Hypoxemia
4	10/29/2022	50	60	Abnormal	Severe hypoxemia
5	10/29/2022	69	97	Normal	Normal
6	10/29/2022	77	97	Normal	Normal
7	10/29/2022	70	97	Normal	Normal
8	10/29/2022	80	205	Normal	Normal
9	10/29/2022	68	96	Normal	Normal
10	10/29/2022	75	215	Normal	Normal
11	10/29/2022	74	96	Normal	Normal
12	10/29/2022	73	204	Normal	Normal
13	10/29/2022	71	97	Normal	Normal
14	10/29/2022	79	204	Normal	Normal
15	10/30/2022	92	206	Normal	Normal
16	10/30/2022	83	97	Normal	Normal
17	10/30/2022	82	60	Normal	Severe hypoxemia
18	10/30/2022	84	96	Normal	Normal
19	10/30/2022	98	191	Normal	Normal
20	10/30/2022	74	97	Normal	Normal

FIGURE 15: Heart rate and oxygen saturation (SpO₂) information on a Google Sheet.

F1-Score: It is the harmonic mean of precision and recall, balancing both measures in a single metric.

$$F1 - \text{Score} = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \times 100. \quad (3)$$

Table 3 provides an overview of fall detection performance in both indoor (carpeted and hard flooring) and outdoor (pavement and grass) test situations. The total number of falls, true positives (TP), false positives (FP), and false negatives (FN) for each scenario are included in the table, which offers information on detection reliability in various contexts. In terms of fall detection, the computed precision, recall, and *F1*-Score show how accurate, sensitive, and effective the system is overall.

3.2.2. Evaluation Metrics of Hand Gesture Communication. The performance of the hand gesture recognition system for emergency message sending is evaluated through two primary metrics: accuracy and average response time.

Accuracy: It assesses the system's ability to correctly recognize gestures. It is calculated as

$$\text{accuracy} = \left(\frac{\text{correct detections}}{\text{total trials}} \right) \times 100. \quad (4)$$

This metric shows the percentage of gestures accurately detected and classified by the system.

Average response time: It measures the time taken by the system to respond to each gesture, recorded in milliseconds (ms). This reflects the system's efficiency in processing and executing the gesture commands.

The effectiveness of the emergency message system with hand gesture recognition is summed up in Table 4. A list of four distinct motions is provided, each of which represents

a distinct emergency function: "Need Food," "Need Water," "Emergency Help," and "Call Attendee." The table shows the average response time, accuracy percentage, total trials, and accurate and wrong detections for each gesture. These outcomes demonstrate how well the system works to consistently identify motions and quickly convey the appropriate emergency message.

3.2.3. Evaluation Metrics. The performance of the SpO₂ and heart rate monitoring system is assessed using three key metrics: average Error, mean absolute error (MAE), and standard deviation.

Average error: It indicates the percentage deviation of device readings, providing a measure of the accuracy of the measurements. It is calculated as

$$\text{average error} = \left(\frac{\text{total error}}{\text{number of readings}} \right) \times 100. \quad (5)$$

MAE: It quantifies the average magnitude of the errors in the readings. It is calculated as

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |\text{Device Reading}_i - \text{Expected Value}_i|. \quad (6)$$

where "n" is the total number of readings.

Standard Deviation: It measures the dispersion of the device readings, indicating how much variation exists from the average reading. It is calculated as

$$\text{standard deviation} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\text{Device Reading}_i - \text{Mean})^2}, \quad (7)$$

where mean is the average of the device readings. A low standard deviation suggests consistent measurements.

TABLE 3: Fall detection performance.

Test scenario	Number of falls	True positives (TP)	False positives (FP)	False negatives (FN)	Precision (%)	Recall (sensitivity) (%)	F1-score (%)
Indoors (carpeted floor)	50	49	1	1	98.0	98.0	98.0
Indoors (hard floor)	50	50	0	0	100.0	100.0	100.0
Outdoors (pavement)	50	49	1	1	98.0	98.0	98.0
Outdoors (grass)	50	48	2	2	96.0	96.0	96.0
Average	—	—	—	—	98.0	98.0	98.0

TABLE 4: Emergency message with hand gesture recognition.

Gesture	Function	Trials	Correct detections	Incorrect detections	Accuracy (%)	Average response time (ms)
Right (need food)	Sends “need food” message	30	29	1	96.7	450
Left (need water)	Sends “need water” message	30	30	0	100.0	440
Forward (emergency help)	Sends “emergency help”	30	30	0	100.0	430
Backward (call attendee)	Sends “call attendee”	30	29	1	96.7	445
Average	—	—	—	—	98.3	441

TABLE 5: SpO₂ and heart rate monitoring performance.

Elderly participant	Metric	Device reading	Average error (%)	Mean absolute error (MAE)	Standard deviation
Elderly 1	SpO ₂ (%)	97	0.9	0.8	0.4
	Heart rate (bpm)	70	1.1	0.8	0.5
Elderly 2	SpO ₂ (%)	96	0.8	0.7	0.4
	Heart rate (bpm)	72	1.0	0.7	0.5
Elderly 3	SpO ₂ (%)	98	0.5	0.5	0.3
	Heart rate (bpm)	68	1.2	0.9	0.6
Elderly 4	SpO ₂ (%)	95	1.1	0.9	0.4
	Heart rate (bpm)	74	1.3	0.9	0.6
Elderly 5	SpO ₂ (%)	97	0.7	0.6	0.4
	Heart rate (bpm)	71	0.9	0.7	0.5
Average	SpO ₂ (%)	96.6	0.683	0.76	0.45
	Heart rate (bpm)	70.4	1.1	0.8	0.54

TABLE 6: Comparison with other works.

No.	Name	Fall detection	Message with hand gesturing	Heart rate	SpO ₂	Data in website	Data in SMS	Data in email	Data in Google Sheet	Data in LCD
1	This paper	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
2	Reference [6]	No	Yes	No	No	No	Yes	Yes	No	No
3	Reference [11]	Yes	No	No	No	No	Yes	No	No	No
4	Reference [13]	Yes	No	No	No	No	Yes	No	No	No
5	Reference [30]	No	No	Yes	Yes	Yes	No	No	No	No
6	Reference [34]	No	No	Yes	No	No	Yes	No	No	Yes

The SpO₂ and heart rate monitoring system performance findings for five senior participants are shown in Table 5. The SpO₂ and heart rate device measurements for each participant are shown in the table together with their corresponding standard deviation, MAE, and average error percentages. The average SpO₂ and heart rate readings are shown in the last row, which represents the monitoring system's overall performance. Together, these parameters evaluate the device's accuracy and dependability in tracking vital signs in senior citizens.

A comparative comparison of the suggested system and a number of previous studies in the fields of vital sign monitoring and fall detection are shown in Table 6. It describes key features such as the ability to detect falls, hand gesture messaging, and heart rate and SpO₂ level monitoring. The chart also lists the different ways that warning data can be transmitted, including via websites, SMS, email, Google Sheets, and LCD screens. The current paper, which is highlighted in the first item, is notable for its thorough integration of vital sign monitoring capabilities, such as SpO₂ and heart rate tracking, with fall detection. In contrast to many previous research studies that concentrate on specific elements, the suggested approach successfully integrates these features into a coherent structure. This method improves user accessibility through a variety of alert channels in addition to increasing the system's overall efficacy. By providing a more comprehensive solution for health monitoring that addresses both fall detection and vital sign assessments in a single system, the proposed study represents a substantial advancement over earlier research.

4. Conclusion and Future Work

The research focused on the development of a smart glove system aimed at enhancing elderly care through real-time monitoring and emergency response features. This study integrated MPU6050 and MAX30100 sensors with Arduino and ESP8266 modules to achieve fall detection, vital signs monitoring (including SpO₂ and heart rate), and emergency communication via hand gestures. Data transmission was facilitated through IFTTT, allowing for alerts and notifications to be sent via SMS, email, and a secured website. The smart glove system addresses critical challenges in elderly care by providing a reliable and comprehensive tool for monitoring and responding to health emergencies. The integration of advanced sensors and real-time communication mechanisms enhances safety and enables timely medical intervention. This system represents a substantial improvement in remote healthcare monitoring, potentially reducing risks and improving the quality of life for elderly individuals. The implementation of this smart glove system marks a significant advancement in elderly care technology. Future research will be essential for further refining the system, expanding its capabilities, and conducting extensive field evaluations to assess its effectiveness in various real-world environments. This ongoing development will contribute to the continued improvement of healthcare solutions for the elderly, addressing their unique needs and enhancing their overall well-being. In future research or developments, it is possible to integrate new algorithms. In upcoming efforts, a variety of

additional sensors can be integrated to measure various health metrics, and alternative microcontrollers may also be considered for implementation.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

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